SVG Configuration Scheme to Enhance Renewable Energy Accommodation Considering Voltage Stability Constraints in Renewable Power Grids

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***Abstract*—This brief proposes an effective Static Var Generator (SVG) configuration scheme to enhance renewable energy accommodation while considering cost efficiency and voltage stability constraints. First, the multiple renewable short-circuit ratio (MRSCR) is employed as a voltage stability indicator, providing a computationally efficient representation of voltage stability constraints. Subsequently, a double-loop multi-objective optimization framework is developed to generate an optimal SVG configuration scheme. The hierarchical structure consists of: (1) an inner-loop optimal power dispatch model that determines the minimum curtailment of renewable energy under both voltage stability constraints and the SVG configuration provided by the outer loop; (2) an outer-loop multi-objective optimization model that simultaneously minimizes SVG configuration costs and annual renewable energy curtailment to establish the SVG configuration. Experimental validation on the SG-118 test system demonstrates the superiority of the proposed framework. Through a comparative analysis of MRSCR characteristics and renewable energy accommodation levels, this brief presents an innovative SVG configuration methodology that effectively balances voltage stability maintenance with renewable energy integration enhancement.**

***Index Terms*—Static var generator configuration, MRSCR, renewable energy accommodation, voltage stability, double-loop multi-objective optimization.**

I. Introduction

UNDER the dual imperatives of environmental sustainability and low-carbon transition, power systems worldwide have witnessed rapid integration of renewable energy (RE) such as photovoltaics (PVs) and wind turbines. While these resources reduce carbon emissions, their inherent intermittency and lack of traditional grid-support functionalities introduce critical challenges to power balance and system stability [1], [2], [3]. These limitations often exacerbate RE curtailment.

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To address RE integration barriers, planning strategies such as large-scale energy storage deployment and coal-fired power plant flexibility retrofits have been widely proposed, primarily targeting intermittency mitigation [4], [5]. However, a less apparent yet increasingly critical constraint lies in the voltage stability of modern power systems–the ability to maintain acceptable voltage levels across all nodes under normal and disturbed conditions. RE-dominated power systems inherently lack the short-circuit capacity and dynamic voltage support provided by synchronous generators, increasing susceptibility to voltage instability [6]. Voltage stability manifests in two forms: static stability and transient stability (overvoltage triggered by faults).

In practice, reactive power compensation devices, such as Static Var Compensator (SVC), Static Synchronous Compensator (STATCOM), Synchronous Condenser(SC), etc, have been used to regulate voltage magnitudes and enhance grid strength by dynamically injecting or absorbing reactive power. Existing research investigated the placement and sizing of reactive power compensation devices to ensure dynamic voltage support during faults by integrating transient overvoltage constraints [7]. Typical approaches include sensitivity indices, intelligent optimization algorithms [8], etc. While current SVG configuration frameworks address voltage stability requirements in operational timescales [9], they neglect a critical dimension: long-term RE curtailment reduction. By focusing solely on improving transient stability for a specific operation condition, these models fail to account for RE output variations and long-term benefits, such as minimized annual RE curtailment. This oversight leads to suboptimal planning outcomes where SVG investments prioritize immediate stability over sustained renewable utilization.

Due to the complex dynamics and nonlinear interdependencies between grid topology, renewable generation, and loads, accurately as well as simply quantifying voltage stability in power systems is inherently complex. The Short-Circuit Ratio (SCR) remains a cornerstone metric for evaluating system strength and voltage stability in numerous relevant studies [10]. Recent advancements extend SCR formulations to address modern grid challenges, such as high penetrations of RE and hybrid AC/DC networks, yielding methods like generalized short-circuit ratio for overvoltage (gSCR-TOV) [11], generalized short circuit ratio (gSCR) for multi-infeed system [12], Site-Dependent SCR (SDSCR) [13], etc. Specifically, Yu et al. [14] demonstrated that the Multiple

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Renewable SCR (MRSCR)—a weighted aggregation of individual renewable plant SCRs—effectively captures voltage sensitivity to power fluctuations in weak grids dominated by inverter-based resources. Similar to reactive power compensation device configurations, a critical gap persists in long-term planning studies, though MRSCR has been applied to short-term operational frameworks like voltage security-constrained unit commitment [15].

In this brief, we propose a double-loop multi-objective optimization framework for configuring Static Var Generators (SVGs) in renewable power systems to address the coupled challenges of voltage instability and RE curtailment in power systems. Section II establishes the foundational analysis for representing voltage stability constraints. Building on this, Section III introduces the double-loop optimization framework. The outer loop formulates a multi-objective optimization problem to minimize SVG configuration costs and annual RE curtailment, while the inner loop solves a stochastic planning-scheduling problem that enforces MRSCR constraints under varying renewable generation scenarios. To validate the framework, Section IV presents a comprehensive case study on the SG118 test system. These outcomes underscore the framework's ability to simultaneously ensure grid stability and maximize renewable utilization, offering a practical pathway for high-penetration renewable integration.

II. Representation of Voltage Stability Constraints

This section elaborates on the critical role of MRSCR in characterizing voltage stability constraints.

A. Voltage Stability Indicator

Voltage stability has emerged as a critical challenge in power systems, significantly constraining the integration of RE. Its representation, however, is inherently complex and computationally intensive. While extensive research has validated the feasibility of modeling static voltage stability, accurately characterizing transient voltage stability remains a significant challenge due to its highly dynamic and nonlinear nature.

MRSCR, defined as the ratio of a renewable node's three-phase short-circuit capacity to its equivalent nodal power, serves as a robust indicator of system strength and voltage stability. This metric effectively captures multi-node interactions, where higher MRSCR values correlate with stronger system strength. The MRSCR of RE node i can be expressed as:

$$R_i(t) = \frac{S_{\mathrm{ac},i}}{P_{\mathrm{RE},i}(t) + \sum_{\substack{j=1 \\ j\neq i}} \left| \frac{Z_{\mathrm{eq},ij}}{Z_{\mathrm{eq},ii}} \right| P_{\mathrm{RE},j}(t)} \tag{1}$$

where $S_{\mathrm{ac},i}$ represents the three-phase short-circuit capacity, $Z_{\mathrm{eq},ij}$ denotes the equivalent impedance between RE nodes i and j, $\frac{Z_{\mathrm{eq},ij}}{Z_{\mathrm{eq},ii}}$ denotes the power conversion factor between multiple RE nodes, and $P_{\mathrm{RE},i}(t)$ is the actual active power output of node i at time t.

TABLE I
Validity Verification of Equation (2)

Node index	Transient voltage rise		MRSCR	
	Weak grid	Strong grid	Weak grid	Strong grid
N-107	0.311	0.290	1.821	1.940
N-108	0.316	0.292	1.734	1.843
N-111	0.304	0.273	2.069	2.191
N-114	0.313	0.289	1.820	1.929
N-120	0.323	0.282	1.790	1.917

B. MRSCR-Based Voltage Stability Constraint

In compliance with international grid codes, post-fault transient voltage rise must remain below 0.3 p.u. So we consider the maximum voltage rise during the post-fault transient process not exceeding 0.3 p.u. as the voltage stability margin. The verification in Table I indicates that enhanced system strength increases the MRSCR, thereby stabilizing the originally unstable nodes. For the system to maintain voltage stability, the following conditions must be satisfied:

$$R_i(t) \geq R_{\mathrm{critical}} \tag{2}$$

where R_{critical} is the critical short circuit ratio.

By combining Equation (1) and (2), we can derive that the equivalent power at the RE nodes must be less than a certain limit. This further constrains the power of RE nodes, thereby affecting the accommodation target.

$$P_{\mathrm{RE},i}(t) + \sum_{\substack{j=1 \\ j\neq i}} \left| \frac{Z_{\mathrm{eq},ij}}{Z_{\mathrm{eq},ii}} \right| P_{\mathrm{RE},j}(t) \leq \frac{S_{\mathrm{ac},i}}{R_{\mathrm{critical}}} \tag{3}$$

The integration of SVGs into power systems effectively enhances MRSCR by reducing the impedance matrix elements $Z_{\mathrm{eq},ij}$ and $Z_{\mathrm{eq},ii}$ and increasing short-circuit current during faults, thereby boosting the short-circuit capacity $S_{\mathrm{ac},i}$. In conclusion, MRSCR effectively characterizes the voltage stability constraints of power systems. The reactive power support capability of SVGs can enhance the MRSCR, thereby strengthening voltage stability and allowing greater integration of RE.

III. Double-Loop Multi-Objective Optimization Framework

In this section, using voltage stability constraints, we propose a double-loop multi-objective optimization framework for configuring SVGs to enhance RE accommodation. As shown in Fig. 1, the framework comprises two loops: The outer loop employs a multi-objective optimization model to minimize SVG configuration costs and annual RE curtailment while providing configuration parameters for the inner loop; The inner loop utilizes an optimal power dispatch model to calculate annual RE curtailment under the outer-loop given SVG configuration, which directly feeds into the outer-loop objectives. Specially, after the dispatch result is given by the inner loop, the objective function 2 of the outer loop is determined accordingly. Therefore, the outer loop will optimize the SVG configuration to achieve the minimization of the RE curtailment in order to minimize the objective function 2.

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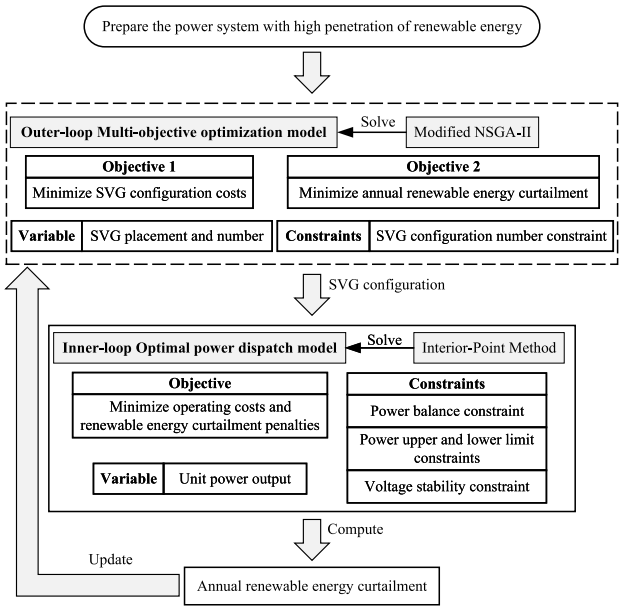


Fig. 1. Double-loop multi-objective optimization framework.

A. Outer-Loop Multi-Objective Optimization Model

1) Objective Functions: The outer loop is formulated as a multi-objective optimization problem with two objective functions.

We consider outer-loop optimization variables to be the number of SVGs configured at different locations:

$$\boldsymbol{x} = [x_1, x_2, \ldots, x_n] \tag{4}$$

where x_i denotes the number of SVGs configured at node i in the power system, and n denotes the number of alternative nodes.

The first objective is to minimize the SVG configuration costs. In practical engineering, the cost can be influenced by multiple factors such as capacity rating, voltage level, geographical location, etc. In this brief, to avoid complicating the mechanism with secondary variables and for simplicity, we assume all configured SVGs have the same capacity and uniform specifications, using a simple quantity–unit price product to represent installation costs:

$$\min \quad f_1(\boldsymbol{x}) = \lambda \sum_{i=1}^{n} x_i \tag{5}$$

where λ is the price for configuring a 50MVar SVG.

Secondly, it is crucial to maintain a high level of RE accommodation. Therefore, the second objective is set to minimize the total annual RE curtailment:

$$\min \quad f_2(\boldsymbol{x}) = \sum_{i=1}^{N_{\mathrm{RE}}} \sum_{t=1}^{T} \left(P_{\mathrm{RE},i}^{\max}(t) - P_{\mathrm{RE},i}(t)\right) \tag{6}$$

where N_{RE} denotes the total number of RE nodes, and T denotes one year encompassing 8,760 hours. $P_{\mathrm{RE},i}^{\max}(t)$ denotes the maximum RE power output of node i at time t.

2) Constraints: In practice, SVGs located near each other electrically can interact, leading to overlapping effects or mutual compensation. To mitigate adverse interactions from multiple SVGs at a single location and simplify the study, we impose a limit L on the number installed per node:

$$0 \le x_i \le L \quad i \in [1, n] \tag{7}$$

B. Inner-Loop Optimal Power Dispatch Model

1) Objective Function: The inner loop is a planning and scheduling problem that arranges the actual RE power generation of each unit at each time. It is optimized with economic efficiency as the objective:

$$\begin{aligned} \min \quad & f(P) = C_1 + C_2 \\ & C_1 = \sum_{i=1}^{N_{\mathrm{g}}} \sum_{t=1}^{T} aP_{\mathrm{g},i}^2(t) + bP_{\mathrm{g},i}(t) + c \\ & C_2 = \sum_{i=1}^{N_{\mathrm{RE}}} \sum_{t=1}^{T} \mu\left(P_{\mathrm{RE},i}^{\max}(t) - P_{\mathrm{RE},i}(t)\right) \end{aligned} \tag{8}$$

where C_1 represents the coal consumption cost of conventional units, while C_2 denotes the penalty term for RE curtailment. N_{g} denotes the total number of conventional units. The coefficients a, b, and c represent the coal consumption characteristics of conventional units. μ is the penalty for RE curtailment by the unit. $P_{\mathrm{g},i}(t)$ denotes the actual conventional unit output of node i at time t.

2) Constraints: The inner scheduling problem mainly considers power balance constraints, upper and lower limits of conventional unit output constraints, RE power output constraints, and additional MRSCR constraints.

Firstly, the normal operation of the power system requires compliance with power balance constraints at each time, that is, power generation always equals power consumption:

$$\sum_{i=1}^{N_{\mathrm{g}}+N_{\mathrm{RE}}} P_i(t) = \sum_{j=1}^{N_{\mathrm{l}}} L_j(t) \tag{9}$$

where N_{l} is the total number of load nodes, and $L_j(t)$ denotes the load power of node j at time t.

Then, for economic benefits, conventional units usually require the generated power to be within the allowable range at each time:

$$P_{\mathrm{g},i}^{\min} \le P_{\mathrm{g},i}(t) \le P_{\mathrm{g},i}^{\max} \tag{10}$$

where $P_{\mathrm{g},i}^{\min}$ and $P_{\mathrm{g},i}^{\max}$ are the lower and upper limits of the output of a conventional unit, respectively. Similarly, the RE power output is always lower than its maximum value:

$$0 \le P_{\mathrm{RE},i}(t) \le P_{\mathrm{RE},i}^{\max}(t) \tag{11}$$

Finally, we add an MRSCR-based constraint to maintain transient voltage stability according to Equation (3).

$$P_{\mathrm{RE},i}(t) \le \frac{S_{\mathrm{ac},i}}{R_{\mathrm{critical}}} - \sum_{j \ne i} \left|\frac{Z_{\mathrm{eq},ij}}{Z_{\mathrm{eq},ii}}\right| P_{\mathrm{RE},j}(t) \tag{12}$$

C. Solution Method

To solve the outer-loop multi-objective optimization model, we adopted the modified NSGA-II (Non-dominated Sorting Genetic Algorithm II) algorithm.

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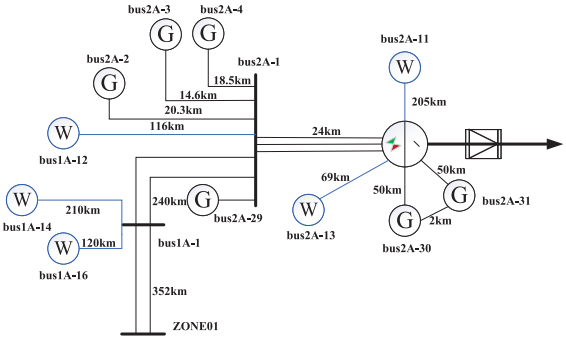


Fig. 2. Topology of the SG118 system ([16]).

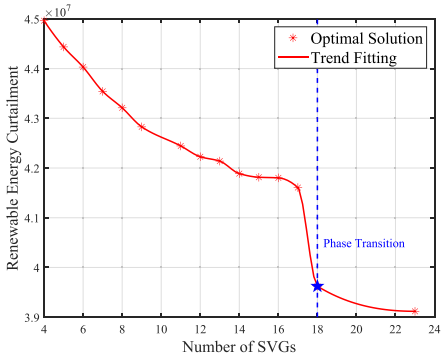


Fig. 3. Pareto optimal solution set.

It can balance two objectives and ultimately generate a set of Pareto front solutions, where each solution is the one that minimizes the other objective while ensuring that one objective is fixed. The inner-loop model belongs to a conventional mixed integer programming problem. Therefore, the optimization problem is solved using the interior-point method by invoking the CPLEX solver on the MATLAB platform.

IV. Case Study

To validate the efficacy of the proposed SVG configuration scheme that deploys the double-loop multi-objective optimization framework, extensive simulations are implemented on the SG118 test power system.

A. Experiment Settings

Fig. 2 shows the topology of the SG118 system. In our simulation, the maximum number L of configurations is set as 5, and R_{critical} is set as 2.

To simplify the calculation process, we cluster the RE scenarios across the entire year using k-means method and obtain the total annual RE curtailment through the weighted calculation of the curtailment under typical scenarios. Therefore, the second objective of the outer loop is changed as:

$$\min \quad f_2(\boldsymbol{x}) = \sum_{k=1}^{K} \epsilon_k A_k(x) \tag{13}$$

where $A_k(x) = \sum_{i=1}^{N_{\text{RE}}} \sum_{t=1}^{24} (P_{\text{RE},i,k}^{\max}(t) - P_{\text{RE},i,k}(t))$, is the RE curtailment for scenario k in one day. ϵ_k is the weight coefficient of scene k, which is the number of days this type of scene appears during a year in the model.

B. Effects of the Proposed Method in Enhancing RE Accommodation

Fig. 3 illustrates the distribution of Pareto front solutions. Each point represents a unique solution balancing competing objectives, specifically the SVG configuration that minimizes RE curtailment for a fixed number of SVGs. A distinct phase transition point is observed at 18 SVGs, indicating a significant shift in the trade-off characteristics. This phase transition point signifies an abrupt change in the system's behavior as parameters reach a critical value.

Fig. 4 illustrates RE accommodation spaces under four typical daily scenarios. The orange-shaded area represents the maximum RE generation potential, while the remaining three regions correspond to operational accommodation ranges for configurations with 0, 5, and 18 SVGs. Notably, the accommodation space is the smallest when no SVGs are configured. As the number of SVGs increases, the feasible accommodation range gradually expands, achieving its largest extent with 18 SVGs. This trend highlights the essential role of SVG configuration in addressing RE variability.

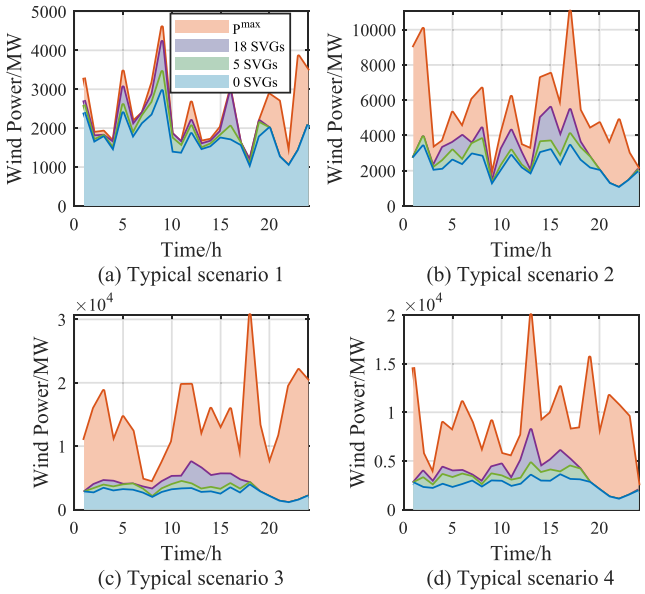


Fig. 4. RE accommodation spaces in four typical daily scenarios with different numbers of SVGs configured.

C. Demonstration of Advantages Over Other SVG Configuration Schemes

We compare annual RE curtailment of our proposed method with two schemes: one with no SVG and a commonly used sequential scheme [17]. The sequential scheme deploys SVGs at nodes with the lowest MRSCR until the predefined limit is reached, considering only RE nodes as candidates. In contrast, our optimized configuration scheme includes collection nodes, which can simultaneously support multiple subordinate RE nodes. As demonstrated in Fig. 5, our optimized configuration scheme achieves a significant reduction in annual RE curtailment.

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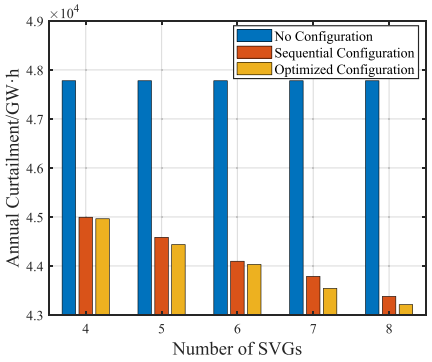


Fig. 5. Comparison of annual RE curtailment.

TABLE II
COMPARISON OF MRSCR USING DIFFERENT METHODS

Comparison Index		None	Sequential	Optimized
Location of SVGs		-	[58, 66, 72, 92, 94]	[7, 73, 82, 90, 94]
Number of SVGs		-	5	5
MRSCR	N-107	1.82	2.04	2.37
	N-108	1.73	5.70	2.11
	N-109	1.88	2.13	2.50
	N-110	1.88	2.12	2.48
	N-114	1.82	5.91	2.10
	N-116	1.98	2.04	6.40
	N-120	1.79	5.47	2.15
	N-121	1.92	1.98	2.14

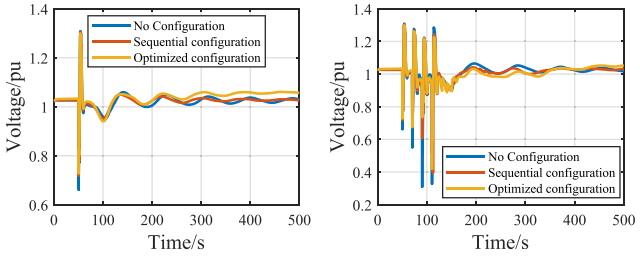


Fig. 6. Comparison of post-fault transient voltage curves with different configurations.

The voltage stability improvement effectiveness of the two configuration schemes both using 5 SVGs is shown in Table II, MRSCR values are specifically listed for nodes that initially violated the MRSCR constraint. Notably, the optimized configuration ensures voltage stability throughout the system, whereas the sequential-configuration scheme fails to meet the critical short-circuit ratio requirement at Node 121, indicating potential voltage instability risks. Compared with their configuration location, optimized configuration scheme configures SVG at some collection nodes. Fig. 6 illustrates transient voltage curves during commutation failure faults across different configurations. The proposed SVG configuration scheme demonstrates superior performance in mitigating both transient overvoltage and undervoltage compared to the baseline.

V. CONCLUSION

This brief proposes a SVG configuration framework that maximizes RE accommodation while ensuring voltage stability at minimal cost. Through a double-loop optimization framework integrating power dispatch and SVG configuration, the method determines the most effective SVG configuration scheme. Simulation results on the SG118 system demonstrate that the proposed solution significantly enhances renewable utilization without compromising voltage stability, offering a practical approach for large-scale RE integration. Future research will include dedicated studies on SVG installation cost modeling and the characterization of mutual interaction effects.

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